

S.P.L.I.T.: Snapping Pendulums Literally In Two

Designed and fully fabricated by engineering & AI students at the University of Prince Mugrin (UPM)

Modern warehouse facilities face a critical spatial challenge, as traditional four-wheeled logistics robots require massive floor space and wide aisles to maneuver safely. To solve this, a team of UPM engineering students fully designed, programmed, and constructed Project S.P.L.I.T., a highly advanced, self-balancing robotic swarm architecture.

By navigating purely on two wheels like an inverted pendulum using PID controllers, these robots maintain a drastically reduced physical footprint. This allows facilities to narrow their aisles and maximize high-density storage without sacrificing payload capacity.

Beyond mechanical stability, the UPM team successfully integrated advanced Artificial Intelligence and computer vision to manage the robotic swarm. The system currently features two distinct operational models to suit different industrial needs:

- **Interactive Gesture Control:** Using an overhead camera system, operators can drive and command the robot in real time using simple hand gestures. This creates a seamless, controller-free environment for human-robot collaboration.
- **Autonomous Path Navigation:** For fully automated environments, the robot utilizes a pre-planned pathfinding model, allowing it to autonomously navigate complex floor plans and execute logistics tasks without human intervention. This path-finding model makes use of surveillance cameras so that no overhead cost is added to the system by using existing essential systems for the purpose of navigation and control.

Project S.P.L.I.T. demonstrates the highest level of multidisciplinary engineering, seamlessly blending mechanical fabrication, complex control theory, and cutting-edge artificial intelligence into a functional, industry-ready prototype.

View this website portfolio for more information: https://abdelrahman-elawady.github.io/portfolio/projects/project_split.html

For full open-source documentation at GitHub: <https://github.com/Abdelrahman-Elawady/SPLIT>

Team members:

- Abdelrahman Elawady 4413746
- Ibrahim Albarbari 4410215
- Ghazi Alhartani 4311147
- Hassan Nasr 4410606